

# A fine-tuned transformer bought +0.03 micro F1. Re-picking thresholds bought +0.11.

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## Results

Four stages — fine-tune, distil, tag, fuse — measured on the eight sealed data/2-eval corpora under the declared policy.

|                                           | cascade_scorecard61      | v5d-fused        | $\Delta$        |
|-------------------------------------------|--------------------------|------------------|-----------------|
| micro F1 ( <i>the ranker</i> )            | 0.7299                   | <b>0.7595</b>    | <b>+0.0296</b>  |
| micro precision                           | 0.6360                   | <b>0.6926</b>    | <b>+0.0566</b>  |
| macro F0.5                                | 0.6230                   | 0.6282           | +0.0052         |
| macro F2                                  | <b>0.6651</b>            | 0.6496           | −0.0155         |
| recall, macro over catalogue              | <b>0.6773</b>            | 0.6367           | −0.0406         |
| worst measurable priority recall          | <b>0.6267</b>            | 0.5881           | −0.0386         |
| priority macro F0.5                       | 0.7980                   | <b>0.8020</b>    | +0.0040         |
| document recall / precision / specificity | 0.7975 / 0.8956 / 0.8792 | <i>identical</i> | —               |
| priority gates passed                     | <b>25/55</b>             | <b>22/55</b>     | −3              |
| one-core p95                              | <b>4.0290 ms</b>         | <b>6.2785 ms</b> | <b>+2.25 ms</b> |

NO FEASIBLE ARM. Nothing promoted.

## The fusion tells you what the content model is worth

fusion chosen per tag: {'and': 37, 'cascade': 24}  
document gate fusion: cascade

Four strategies were available per tag — cascade, content, or, and — each scored by F0.5 on the calibration carve. The result is unambiguous:

- content **was never chosen**. Not once in 61 tags. There is no tag the content model answers better on its own than the cascade does.
- or **was never chosen**. The content model finds nothing the cascade misses; if it did, or would have won somewhere.
- and **won 37 times** — an intersection, which can only *remove* firings. The content model's entire contribution is as a **precision veto**.
- **The document gate stayed the cascade's**, so the customer-facing question — does this document contain PII — was not improved at all.

That is consistent with every number above: precision up, recall down, F2 down.

The comparison that matters

The same day, on the same corpora and evaluator, re-selecting the cascade's 61 thresholds into a (precision, recall) box — no training, no new data, no extra latency — produced:

|                                  | micro F1 | micro precision | p95      |
|----------------------------------|----------|-----------------|----------|
| cascade_scorecard61              | 0.7299   | 0.6360          | 4.029 ms |
| v5d-fused (this run)             | 0.7595   | 0.6926          | 6.279 ms |
| cascade_p80r70 (thresholds only) | 0.8394   | 0.9165          | 4.029 ms |

The threshold re-selection beat the entire transformer chain by 0.08 micro F1 and 0.22 micro precision, at zero latency cost and zero training. The chain cost a GPU fine-tune over 451,548 documents, a distillation, 58 minutes of head fitting, and 2.25 ms of every future inference, to arrive at less than a third of the gain.

TL;DR

- **The chain works end to end and passes its serving constraint:** measured fused p95 **6.279 ms** against the 8 ms requirement.
- **Every stage met its checkpoint.** v5a improved 50 of 61 tags (mean +0.41); v5b's static table separates identifier tokens better than the span-stage table did (probe cosine 0.0848); v5c selected all 61 thresholds.
- **And the result is still small.** +0.0296 micro F1, -3 priority gates, +56% latency.
- **The content model is a precision veto and nothing else** — and on 37 tags, content and or on none, and no contribution to the document gate.
- **Re-picking thresholds was ~4× more valuable**, free, and instant.
- **The premise was reasonable and the measurement refutes it.** The hypothesis was that contextual signal would find identifiers a bag of hashed n-grams cannot. The fusion says the cascade was already finding them; what it lacked was knowing when to stay quiet, and a threshold answers that far more cheaply than a transformer.

|          |                                                                                                                                                                                                                                   |
|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Date     | 2026-08-26                                                                                                                                                                                                                        |
| Project  | projects/pii-content-v5                                                                                                                                                                                                           |
| Run      | 26-08-26-1                                                                                                                                                                                                                        |
| Request  | "finetune the LLM kalyan-ks/ettin-68m-nemotron-pii on all 8 training datasets, model to vec it with token based not sentence based, then train that model on the same 8 training sets, then test adding that model to this model" |
| Scope    | 4 stages; 1 fused arm × 8 sealed corpora; 61 labels                                                                                                                                                                               |
| Hardware | RTX 4080 (12 GB) for v5a/v5b; 1 CPU core for every published latency                                                                                                                                                              |

|             |                                                                                              |
|-------------|----------------------------------------------------------------------------------------------|
| Feasibility | recorded unknown before the budget — no ceiling existed for the fused metric                 |
| Outcome     | chain completed, serving constraint met, <b>blocked</b> by the recall gate, nothing promoted |

## Model scoreboard

| Model                        | Dataset                                   | F1 @5 | F1 @3 | F1 @1 | Macro F1 | Docs/sec | ms/doc |
|------------------------------|-------------------------------------------|-------|-------|-------|----------|----------|--------|
| cascade-scorecard61-baseline | 10360_betterdataai_ner_silver_eval_10.36k | 0.225 | 0.180 | 0.126 | 0.263    | 159.3    | 6.3    |
| cascade-scorecard61-baseline | 10626_ai4privacy_pii_masking_eval_10.63k  | 0.758 | 0.711 | 0.611 | 0.715    | 159.3    | 6.3    |
| cascade-scorecard61-baseline | 20000_pii_holdout_20.00k                  | 0.955 | 0.947 | 0.922 | 0.739    | 159.3    | 6.3    |
| cascade-scorecard61-baseline | 30000_pii2_eval_25.15k                    | 0.926 | 0.911 | 0.862 | 0.761    | 159.3    | 6.3    |
| cascade-scorecard61-baseline | 38937_openpii_pii_eval_38.94k             | 0.992 | 0.988 | 0.979 | 0.692    | 159.3    | 6.3    |
| cascade-scorecard61-baseline | 4000_datax-dualjudge-evalset-1.32k        | 0.552 | 0.463 | 0.311 |          | 159.3    | 6.3    |
| cascade-scorecard61-baseline | 5617_nemotron_eval_5.36k                  |       |       |       |          | 159.3    | 6.3    |
| cascade-scorecard61-baseline | 6589_govdocs2-dualjudge-eval20-3.53k      | 0.578 | 0.542 | 0.471 |          | 159.3    | 6.3    |
| pii-content-v5d-fused        | 10360_betterdataai_ner_silver_eval_10.36k | 0.225 | 0.180 | 0.126 | 0.245    | 159.3    | 6.3    |
| pii-content-v5d-fused        | 10626_ai4privacy_pii_masking_eval_10.63k  | 0.758 | 0.711 | 0.611 | 0.711    | 159.3    | 6.3    |
| pii-content-v5d-fused        | 20000_pii_holdout_20.00k                  | 0.955 | 0.947 | 0.923 | 0.727    | 159.3    | 6.3    |
| pii-content-v5d-fused        | 30000_pii2_eval_25.15k                    | 0.926 | 0.911 | 0.862 | 0.776    | 159.3    | 6.3    |
| pii-content-v5d-fused        | 38937_openpii_pii_eval_38.94k             | 0.992 | 0.988 | 0.979 | 0.694    | 159.3    | 6.3    |
| pii-content-v5d-fused        | 4000_datax-dualjudge-evalset-1.32k        | 0.552 | 0.462 | 0.311 |          | 159.3    | 6.3    |
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Measured 2026-08-26 · n=10,360, n=10,626, n=20,000, n=30,000, n=38,937, n=4,000, n=5,617, n=6,589.